

Package ‘abpool’

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Title Approximate Bayesian Pooling for Multiple Imputation

Version 0.0.0.9000

Description Implements approximate Bayesian pooling (ABpool) for statistical inference after multiple imputation as described in Phillips, Christodoulou and Steinsaltz, ``Pooling after multiple imputation under posterior skewness" (in preparation). Provides functions to generate approximate posterior samples, calculate credible intervals, and a diagnostic for Rubin's rules.

Imports mice,
mvtnorm

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URL <https://github.com/danphillipsstats/abpool>

BugReports <https://github.com/danphillipsstats/abpool/issues>

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abpool	<i>Approximate Bayesian pooling</i>
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Description

Perform approximate Bayesian pooling (ABpool) on fitted models after multiple imputation.

Usage

```
abpool(object, parameters = NULL, dfcom = NULL, J = 1)
```

Arguments

object	Either a mira object, or a list of fitted model objects, where the <i>l</i> th element gives the model fit to the <i>l</i> th imputed dataset.
parameters	A character vector of parameter names, or a numeric vector of parameter positions to include in ABpool. If <code>parameters = NULL</code> , ABpool will be performed for all parameters in the model. The parameters are included in the order specified.
dfcom	The complete-data degrees of freedom used for the t-distributed complete-data posterior approximation. <code>dfcom = Inf</code> corresponds to a Gaussian approximation. If <code>NULL</code> , <code>abpool()</code> attempts to extract the degrees of freedom from each fitted model using <code>df.residual()</code> ; an error is returned if extraction fails or the extracted values differ across imputations. We recommend specifying <code>dfcom</code> when the appropriate complete-data degrees of freedom are known.
J	Number of samples drawn per imputed dataset. Defaults to 1.

Details

Given model fits from multiply imputed datasets, `abpool()` generates posterior samples using approximate Bayesian pooling (ABpool). The function first extracts estimates and variances for the parameters from each fitted model, before generating ABpool posterior samples.

The `abpool()` function samples from an approximation to the observed-data posterior distribution, assuming the completed-data posterior distribution given each imputed dataset is t-distributed, with degrees of freedom equal to `dfcom`, location given by the estimate, and squared scale (or scale matrix) given by the variance estimate. For `dfcom = Inf`, draws are from the corresponding Gaussian distribution. The input object may be:

1. A list of model fit objects, with each element generated from a function such as `lm()`, `glm()`, `survival::coxph()`.
2. An object of class `mira` generated by applying `with()` to a `mids` object from the `mice` package. Suitable `coef()` and `vcov()` methods must be available for the fitted models. Classes supported include outputs from `lm()`, `glm()` and `survival::coxph()`. Note `dfcom` must be user-supplied for `coxph()` models.

Inference using ABpool generally requires substantially more imputations than Rubin's rules. We recommend using at least 200 imputations when comparing Rubin's rules with ABpool using a Q-Q plot (`QQ_compare_abpool_rubin()`). Even larger numbers of imputations may be required for final inference using ABpool to achieve appropriate coverage (Phillips, Christodoulou and Steinsaltz, 2026).

Value

An object of class `abpool`. The object is a list containing

samples ABpool posterior samples. A numeric vector of length $m * J$ for a single parameter, or an $m * J$ by p matrix for p parameters.

estimates Parameter estimates extracted from the fitted models. A numeric vector of length m for a single parameter, or a list of m numeric vectors of length p for multiple parameters.

variances Variance estimates extracted from the fitted models. A numeric vector of length m for a single parameter, or a list of m variance-covariance matrices of dimension p by p for multiple parameters.

dfcom The complete-data degrees of freedom used for sampling.

parameters A character vector containing the names of the parameters included in ABpool, in the order used for sampling.

imputation An integer vector of length $m * J$ identifying the imputed dataset associated with each element (scalar case) or row (multivariate case) of samples.

J The number of samples drawn per imputed dataset.

m The number of imputed datasets.

References

Phillips, Christodoulou and Steinsaltz, *In preparation*. Pooling after multiple imputation under posterior skewness.

See Also

`qqplot_rubin_abpool()` to compare Rubin's rules and ABpool in a quantile-quantile (Q-Q) plot; `confint.abpool()` to generate ABpool credible intervals; `abpool_sample()` to sample from a vector/list of estimates and variances.

Examples

```
# Generate data
set.seed(1)
n <- 400; nobs <- 40
X <- rnorm(n); Z <- sqrt(0.9)*rnorm(n) + sqrt(0.1)*X
beta_0 <- 0; beta_X <- 0.8; beta_Z <- 0.2
eta <- beta_0 + X*beta_X + Z*beta_Z
Y <- rbinom(n, size = 1, p = exp(eta)/(1 + exp(eta)) )
X[1:(n-nobs)] <- NA
log.data <- data.frame(Y = Y, X = X, Z = Z)
m <- 200
# Impute using mice (200 imputations, predictive mean matching)
imp <- mice::mice(log.data, m = m, method = "pmm", print = FALSE)
fits <- with(
  imp,
  glm(Y ~ X + Z, family = binomial)
)
# Fit ABpool
abpool(fits)
```

abpool_sample	<i>abpool_sample</i>
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Description

Draw samples via approximate Bayesian pooling (ABpool).

Usage

```
abpool_sample(estimates, variances, dfcom, J = 1)
```

Arguments

<code>estimates</code>	Estimates from the imputed datasets.
<code>variances</code>	Associated variances or variance-covariance matrices for the estimates from the imputed datasets.
<code>dfcom</code>	Complete-data degrees of freedom.
<code>J</code>	Number of samples per imputation.

Details

For each imputed dataset, sample from a t-distribution with location given by `estimates`, and scale given by `variances`. For `dfcom = Inf`, draws are from the corresponding Gaussian distribution.

Value

A numeric vector of posterior draws for a scalar parameter, or a matrix of posterior draws for multiple parameters.

References

Phillips, Christodoulou and Steinsaltz, *In preparation*. Pooling after multiple imputation under posterior skewness.

See Also

[abpool\(\)](#) to sample from a `mira` object or list of fitted models. [abpool\(\)](#) also outputs an object which can be analysed by the rest of the package e.g. [qqplot_rubin_abpool\(\)](#) and [confint.abpool\(\)](#), unlike [abpool_sample\(\)](#).

Examples

```
# Scalar input
estimates <- c(1,2,3)
variances <- c(1,2,3)
abpool_sample(estimates, variances, dfcom = Inf)

# Multi-parameter input
estimates <- list(
  c(x = 1, z = 2),
  c(x = 3, z = 4)
)
```

```

variances <- list(
  matrix(c(1,0.1,0.1,2), nrow = 2, ncol=2),
  matrix(c(1,0,0,1), nrow = 2, ncol=2)
)
abpool_sample(estimates, variances, dfcom = 19)

# Example with manually generated model fits after multiple imputation
# Generate data
set.seed(1)
n <- 400; nobis <- 40
X <- rnorm(n); Z <- sqrt(0.9)*rnorm(n) + sqrt(0.1)*X
beta_0 <- 0; beta_X <- 0.8; beta_Z <- 0.2
eta <- beta_0 + X*beta_X + Z*beta_Z
Y <- rbinom(n, size = 1, p = exp(eta)/(1 + exp(eta)) )
X[1:(n-nobis)] <- NA
log.data <- data.frame(Y = Y, X = X, Z = Z)
m <- 200
# Illustrative imputation procedure
estimates <- list(); variances <- list()
for (i in 1:m){
  X[1:(n-nobis)] <- rnorm(n-nobis)
  log.data.imp.i <- data.frame(Y = Y, X = X, Z = Z)
  glm.out.i <- glm(Y ~ X + Z, family = binomial)
  estimates[[i]] <- coef(glm.out.i)
  variances[[i]] <- vcov(glm.out.i)
}
# Sample from ABpool
abpool_sample(estimates,variances, dfcom = Inf)

```

confint.abpool

Credible intervals for approximate Bayesian pooling

Description

Calculate credible intervals using posterior samples from approximate Bayesian pooling (ABpool).

Usage

```

## S3 method for class 'abpool'
confint(object, parm = NULL, level = 0.95, ...)

```

Arguments

object	An object of class abpool, returned by abpool() .
parm	A character or numeric vector specifying the parameters for which credible intervals should be calculated. If NULL, intervals are calculated for all parameters.
level	Credible interval level. A single numeric value between 0 and 1. Defaults to 0.95.
...	Additional arguments. Currently unused.

Details

Credible intervals are calculated using empirical quantiles of the posterior samples returned by `abpool()`.

Value

A matrix containing the lower and upper credible limits for each requested parameter, with one row per parameter.

See Also

`abpool()`, `qqplot_rubin_abpool()`

Examples

```
# Generate data
set.seed(1)
n <- 400; nobs <- 40
X <- rnorm(n); Z <- sqrt(0.9)*rnorm(n) + sqrt(0.1)*X
beta_0 <- 0; beta_X <- 0.8; beta_Z <- 0.2
eta <- beta_0 + X*beta_X + Z*beta_Z
Y <- rbinom(n, size = 1, p = exp(eta)/(1 + exp(eta)) )
X[1:(n-nobs)] <- NA
log.data <- data.frame(Y = Y, X = X, Z = Z)
m <- 200
# Impute using mice (200 imputations, predictive mean matching)
imp <- mice::mice(log.data, m = m, method = "pmm", print = FALSE)
fits <- with(
  imp,
  glm(Y ~ X + Z, family = binomial)
)
# Fit ABpool
abpool.out <- abpool(fits)
confint(abpool.out)
```

<code>qqplot_rubin_abpool</code>	<i>Quantile-Quantile plot comparing Rubin's rules with ABpool</i>
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Description

Generates a Q–Q plot comparing the quantiles from Rubin's t-approximation with posterior samples generated by approximate Bayesian pooling (ABpool).

Usage

```
qqplot_rubin_abpool(
  object,
  parm = NULL,
  plot.it = TRUE,
  add_qqline = TRUE,
  ...
)
```

Arguments

<code>object</code>	An object of class <code>abpool</code> , produced by <code>abpool()</code> .
<code>parm</code>	A character or numeric vector specifying the parameters for which to generate the Q–Q plot. If <code>NULL</code> , a Q–Q plot is generated for every parameter in <code>object</code> .
<code>plot.it</code>	Logical. Should the Q–Q plot be plotted? Defaults to <code>TRUE</code> .
<code>add_qqline</code>	Logical. Should a Q–Q reference line be added? Defaults to <code>TRUE</code> .
<code>...</code>	Additional graphical arguments passed to <code>qqplot()</code> .

Details

The plot is intended to diagnose potential discrepancies in the distribution given by Rubin’s rules by comparing it with the corresponding distribution obtained from ABpool.

The quantile probabilities are generated using `ppoints()`, in a manner analogous to `qqplot()` and `qqnorm()`. Quantiles from Rubin’s t-approximation are plotted on the x-axis and the ABpool posterior samples on the y-axis.

For each parameter, Rubin’s t-approximation has location $\bar{\theta} = \frac{1}{m} \sum_{l=1}^m \hat{\theta}^{(l)}$ and squared scale $V = \bar{U} + (1 + \frac{1}{m}) B$, where $\bar{U} = \frac{1}{m} \sum_{l=1}^m U^{(l)}$ is the mean within-imputation variance and $B = \frac{1}{m} \sum_{l=1}^m (\hat{\theta}^{(l)} - \bar{\theta})^2$ is the between-imputation variance. The degrees of freedom are calculated using the Barnard–Rubin method, which simplifies to Rubin’s original method when the complete-data degrees of freedom are infinite. The complete-data degrees of freedom are chosen to match those used by ABpool.

The resulting t-distribution is compared with the ABpool posterior samples by comparing ordered samples to t-quantiles, with the number of quantiles equal to the number of ABpool samples.

Value

Invisibly returns the following: If `parm` specifies a single parameter, a list containing:

- x** The quantiles of Rubin’s t-approximation at the probabilities given by `p`. These are the x-coordinates of the points plotted, or that would be plotted if `plot.it = FALSE`.
- y** The ABpool posterior samples, sorted in increasing order. These are the y-coordinates of the points plotted, or that would be plotted if `plot.it = FALSE`.
- p** The probabilities used to calculate the quantiles of Rubin’s t-approximation used in the plot, generated using `ppoints()`.

If multiple parameters are specified, a named list containing one such list for each parameter is returned.

References

Phillips DJ, Christodoulou MD and Steinsaltz D, *In preparation*. Pooling after multiple imputation under posterior skewness.

Barnard, J and Rubin, DB (1999). Small-sample degrees of freedom with multiple imputation. *Biometrika*, 86(4), 948–955.

Rubin, DB (1987). *Multiple Imputation for Nonresponse in Surveys*. Wiley.

See Also

`abpool()`, `confint.abpool()`

Examples

```
# Generate data
set.seed(1)
n <- 400; nobs <- 40
X <- rnorm(n); Z <- sqrt(0.9)*rnorm(n) + sqrt(0.1)*X
beta_0 <- 0; beta_X <- 0.8; beta_Z <- 0.2
eta <- beta_0 + X*beta_X + Z*beta_Z
Y <- rbinom(n, size = 1, p = exp(eta)/(1 + exp(eta)) )
X[1:(n-nobs)] <- NA
log.data <- data.frame(Y = Y, X = X, Z = Z)
m <- 200
# Impute using mice (200 imputations, predictive mean matching)
imp <- mice::mice(log.data, m = m, method = "pmm", print = FALSE)
fits <- with(
  imp,
  glm(Y ~ X + Z, family = binomial)
)
# Fit ABpool
abpool.out <- abpool(fits)
# Compare ABpool and mice in a QQ plot
qqplot_rubin_abpool(abpool.out, parm = "X")
```

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